

Standard Form — Writing and Converting

AQA · KS4 Year 9–10

Answer each question in your workbook. Show your working.

COMMON MISTAKES

A must satisfy $1 \leq A < 10$. So 36×10^4 is wrong; correct: 3.6×10^5 .

Positive power = big; negative power = small.

Count places carefully — 0.00047 needs 4 moves → 4.7×10^{-4} .

$0.003 = 3 \times 10^{-3}$, NOT 3×10^3 .

METHOD

- 1 Move decimal so $1 \leq A < 10$
- 2 Count places moved = power n
- 3 Big number (≥ 1) → positive n
- 4 Small number (< 1) → negative n
- 5 Check by converting back

WORKED EXAMPLE

Write 0.000 47 in standard form.

Step 1 — Move decimal 4 places: 4.7

Step 2 — Original < 1 , so n = -4

Answer: 4.7×10^{-4}

VISUAL AID

Place-value slide showing the decimal point moving left for big numbers (positive power) and right for small numbers (negative power). Arrow diagram: 5,600,000 → 5.6 (6 places left) = 5.6×10^6 . And 0.0034 → 3.4 (3 places right) = 3.4×10^{-3} .

Standard form: a) 5,000 b) 340,000 c) 72,000,000 d) 810

Standard form: a) 0.003 b) 0.00056 c) 0.0000012 d) 0.47

To ordinary: a) 3.2×10^4 b) 7.05×10^6 c) 1.8×10^{-3} d) 9.1×10^{-5}

Not standard form? Fix: a) 5.6×10^3 b) 36×10^4 c) 0.8×10^{-2} d) 7×10^9

[illegible]

Earth-Sun = 1.5×10^8 km; Earth-Mars = 2.3×10^8 km. a) Sun distance ordinary. b) Closer? c) Difference in standard form.

Order smallest first: 3.2×10^{-3} , 8.1×10^{-4} , 5.5×10^{-3} , 2.9×10^{-2} . Strategy?

Bacterium = 2×10^{-6} m; virus = 3×10^{-8} m. a) How many times longer? b) 500 bacteria end-to-end? c) Viruses in 1 mm?